

Mini-Project Final Technical Memo

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IED Section: 5

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Introduction

The main objective of this project is to attempt to achieve an automated car that can follow a black line width of $\frac{3}{4}$ th of an inch on a white background while being both cost-efficient and reliable. The developed car must work reliably well and will not have unexpected issues randomly, meaning that its performance is not dependent on luck and should adapt to various testing conditions. The car must be easy and obvious to assemble, taking no less than 10 minutes for even an amateur to put together. The car must also be cost-efficient, built with inexpensive components that cost less than thirty dollars, and easily stored. The components include motors, wheels, chassis, which acts as the structure for the car, a motor driver used to control the speed and direction of the car, two ground-facing infrared sensors(IR sensors) to detect the car's position, as well as an Arduino microprocessor used to control how the car is going to behave through the motor driver based on the results from the IR sensors. All of these are powered by one 9-volt battery. This car is designed with an educational purpose in mind. With the combination of low cost and ease of assembly, teachers will not have much trouble purchasing one while students can gain hands-on experience from assembling the car from scratch and can also alter or improve the design of the vehicle at their curiosity to further or strengthen their knowledge.

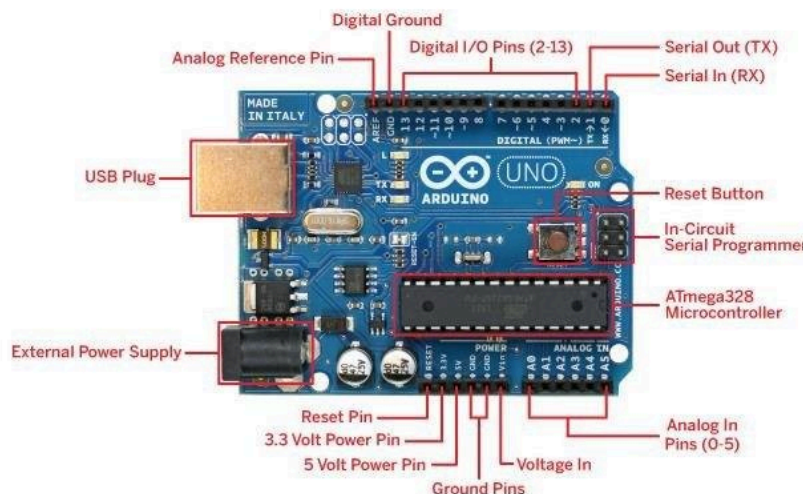
Concepts and Benchmarking

Figure 1: Amazon listings for two different line following kits with different components, the left costs 60 dollars while the right costs 17 dollars.



After some investigation and research into existing line follower technology and products, numerous line follower robot cars are sold along with a soldering kit as a fun and educational learning experience to teach youth about soldering with a price ranging from 10 dollars upwards to 70 dollars. These robot cars typically differ from each other in their respective motor technology and their sensing unit causing the prices to fluctuate depending on which component is used. For example, in Figure 1, the ELEGOO UNO R3 Smart Robot Car costs sixty dollars because it uses a camera instead of photoresistors and four motors instead of two when compared to the 17 dollar 2 Pack D2-5 Smart Car Kit Soldering Project Line Following Robot on the right of Figure 1, even with all the extra tools and resources provided by the ELEGOO UNO kit, it can be seen how the price can fluctuate with better components.

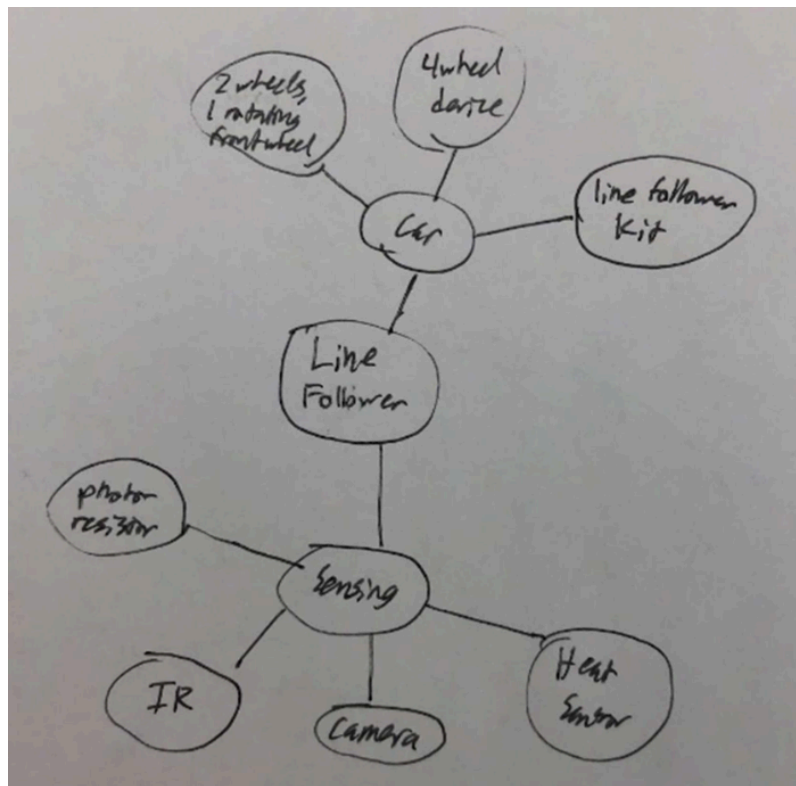
Figure 2: Arduino board and pin configuration in detail



The three main concepts for the line follower robot are the motor unit, the sensing unit, and the control unit. The control unit is the least contested concept as its purpose is to process the

information sent from the sensing unit to make decisions like turning left, right, or straight. Microcontrollers like the Arduino in Figure 2, are tiny circuits that can be programmed to respond and execute instructions on devices and are used as the main control unit for the line follower. They are small chips that contain a processor, memory for storing data and instructions, and input/output pins to connect to other devices. The Arduino is very simple and accessible, making it the ideal tool for beginners to learn or experiment with coding and electronics, it is also suitable for creating an educational line follower model because of how accessible and easy to use it is.

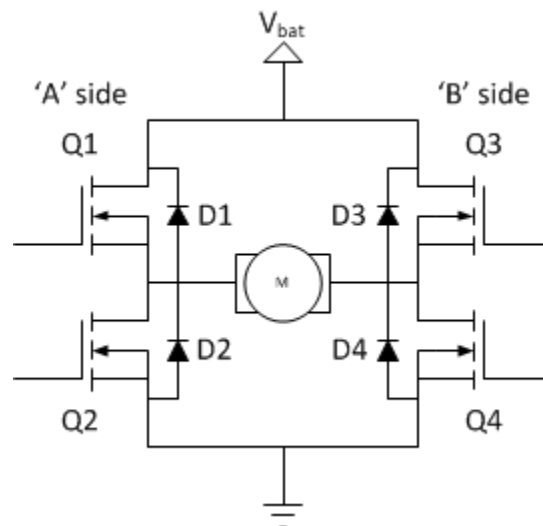
Figure 3: Mind Map



The motor unit, as seen in Figure 3, is the top portion of the mind map and an important part of the design of the line follower robot. The crucial part is that the final design must be able to handle sharp turns like 90-degree turns without failure, meaning that the design requires an H-bridge because this specific integrated circuit would allow the motor to spin forward or backward. The H-bridge is made with four transistor switches lined up in an H formation as shown in Figure 4 that turn on and off to control both the speed and the direction. To control the

direction, the H-bridge will change the polarity of the voltage applied to the motors and by allowing current to flow through specific switches the direction can be changed. To control the speed, the H-bridge rapidly turns the switches on and off to vary the voltage applied, thereby averaging the voltage applied to control the speed, this process is called pulse width modulation. The two options are to either build a transistor H-bridge from scratch with the downside of a potential failure due to inaccuracies or mistakes making it or buy an L298N motor driver, which essentially is a dual full-bridge driver designed to control two DC motors containing two H-bridge circuits within a single package, allowing it to control the direction and speed of two separate motors independently but costing money and is less efficient.

Figure 4: H-bridge Schematic



The sensing unit, as shown in Figure 3, is the bottom portion of the mind map and detects whether or not the line follower robot is on the right track or adjustments are needed. Sensors like photoresistors, IR sensors, and even cameras have always been a part of any line follower robots, as shown in Figure 5 because the robot needs a way to detect turns and intersections. These sensing units pick up on information about the road ahead and relay the information to the control unit which will then decide a course of action for the motor unit. Photoresistors's resistance changes depending on the amount of light they receive, so placing them over a black line will increase their resistance and using a voltage divider to measure if the voltage across the photoresistor exceeds or fails to exceed a certain threshold. The problem with this approach is that the performance of the photoresistors depends on the lighting of the room, well-lit rooms or

dimly lit rooms could have an impact on the threshold thus causing errors. IR sensors, or infrared sensors, emit infrared light onto a surface to capture the reflected light and output a voltage that corresponds to the reflection's intensity. In this specific scenario, the black surface absorbs more light than the white surface meaning that the output voltage will be lower when the sensor detects black. The problem with this approach is that the IR sensors have a limited range of a few centimeters and often cost more than the photoresistor. For the camera, an image analyzation algorithm will be needed to analyze the camera's feed pixel by pixel to determine if the line follower robot is on the right path or not, if not then adjustments have to be made based on the feed. This method eliminates most inaccuracies but is far more complex than using photoresistors or ID sensors, not to mention more expensive since the camera is way more expensive than IR sensors and photoresistors.

Figure 5: Camera(Left), IR sensor(Middle), and Photoresistor(Right)

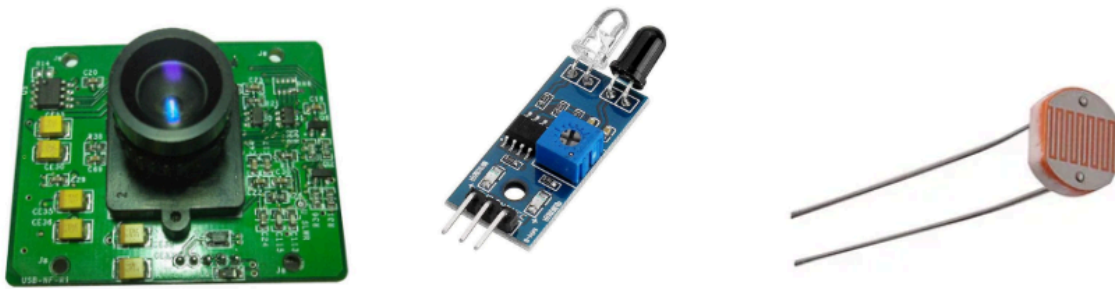


Table 1: Concept Selection Matrix

Selection Criteria	Self-made transistor H-bridge	L298N Motor Driver	Photoresistor	Camera	IR-sensor
Cost	1	1	1	-1	0
Difficulty of Implementation	-1	1	0	-1	1
Efficiency	1	0	0	0	1
Creativity	1	0	-1	1	-1
Safety	0	1	1	1	1
Score	2	3	1	0	2
Continue?	n	y	n	n	y

For concept Selection, a concept selection matrix is implemented to select the component that best fits the criteria shown in Table 1 for both the motor unit and the sensing unit, separated by the thick vertical line. Scoring a 1 indicates a strong performance and 0 indicates a slightly worse performance, while -1 means the component would have a negative impact because of its poor performance. If two parts are around the same, then the two would both be given 1s. In the end, the ones with the highest net score are picked to be implemented. The first iteration uses the L298N motor driver shown in Figure 6 along with the IR sensors shown in Figure 7 attached to the chassis at a slanted angle with the microprocessor Arduino between the motor driver and sensors as shown in Figure 8.

Figure 6: Sketch of the Motor Unit of the L298N motor-driver

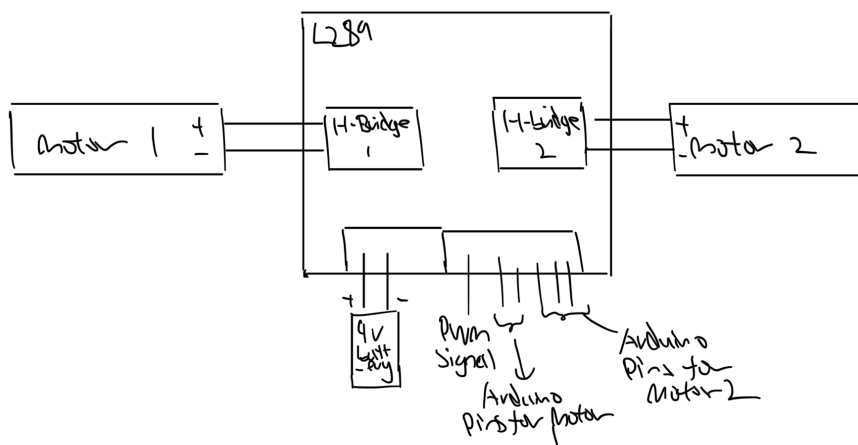


Figure 7: Sketch of the Sensing Unit the IR sensors

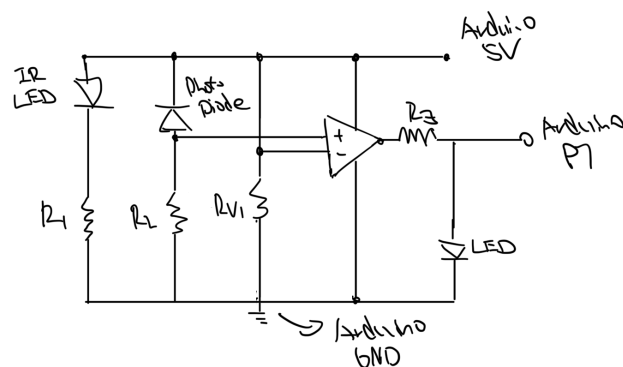
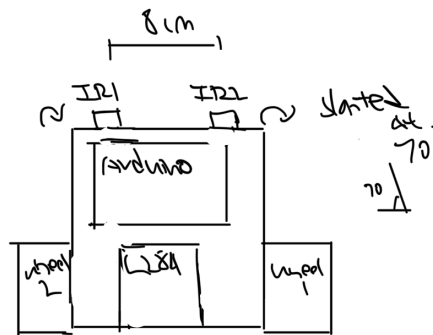
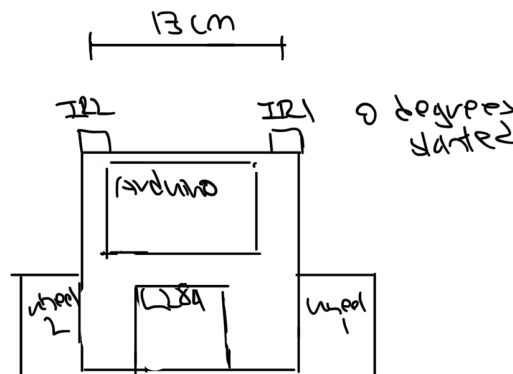


Figure 8: Sketch of the first iteration design



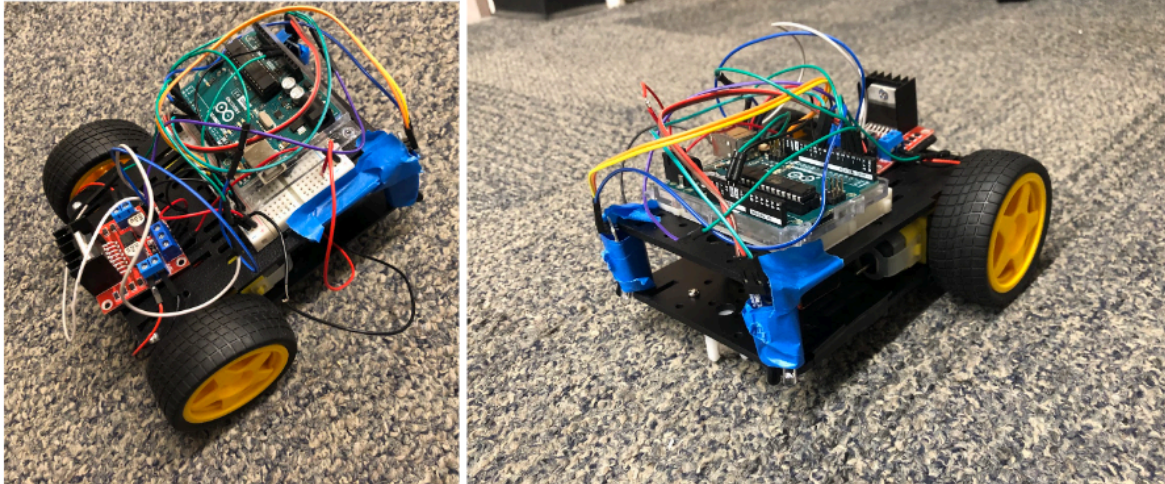
However, after further testing, it seems that the IR sensors are not able to detect the ground as accurately as they should have been and would often cause the car to drive off the track as it is not able to pick up on the black line on the test boards. To fix this error, the IR sensors are shifted more towards the ground and are no longer slanted but directly pointing downwards. This change helped the IR sensors detect the line better and improved the results of the line follower robot drastically, but due to time constraints and the looming competition date, this was the last iteration of the robot which is shown below in Figure 9.

Figure 9: Final Iteration of the Line Follower Robot



Solution

Figure 10: Competition day line follower robot

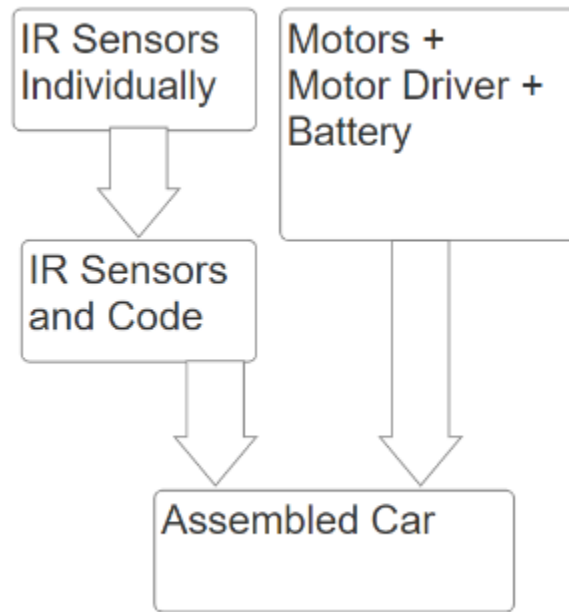


The final design is made using the Arduino microprocessor for the control unit, two IR sensors for the sensing unit, and one L298N motor driver as the motor unit. The two IR sensors are placed 13 centimeters away from each other and directly point down on the ground. The Arduino sat on top of the breadboard with the L298N motor driver placed at the back of the robot. The motors are connected to the L298N and placed on the bottom of the chassis along with the one 9-volt battery used to power the car. The line follower robot begins to move once the positive wire of the 9-volt battery is connected to the breadboard which is also connected to both the Arduino and motor driver. The photo of the final design is shown in Figure 10 and the sketch in Figure 9.

Test Plan and Test Results

The approach for testing the line follower robot was to test the minimal individual building blocks first, then after combining any two blocks, test that they work together. That was testing the IR sensors and the power system in parallel. The power system consists of the motors, the L298N motor driver, and the 9-volt battery. Following that, testing the Arduino code can properly detect the IR sensor's output and correct logic can be obtained. Then finally the entire assembled car. This testing flow chart is shown in Figure 11.

Figure 11: Testing Flow Chart



The IR sensors had two LEDs. One of the LEDs would be on if it was powered, and the other LED would be on if the IR sensor saw its infrared reflected back. From now on, the former LED will be referred to as the power LED, and the result LED for the latter. In this case, the result LED would be on for white paper, and off for black lines. The testing consisted of checking to see if the IR sensors would properly detect the white on a white piece of paper and 3 shades of black; printed black, black tape, and marker black. As seen in Table 2, the result LED works as expected and the IR sensors are confirmed to function.

Table 2: Individual IR sensor testing results table

	Power LED	Result LED	Desired result LED
White paper	On	On	On
Printer black	On	Off	Off
Black tape	On	Off	Off
Marker Black	On	Off	Off

The power system consisted of using the 9-volt battery fed into the motor driver's power pin, and the speed control pins, with the motor driver connected to both motors. Testing consisted of plugging and unplugging the corresponding wire for both motor's motor driver input pins individually to see if they would spin in the correct direction if plugged into power or not. Looking at Table 3, each motor turns on when the corresponding input pin on the motor driver is given power, so the power system and motor components are functional.

Table 3: Motor, motor driver, and battery testing results table

Left motor input	Right motor input	Left motor	Right motor	Driving direction
0V	0V	Off	Off	Stationary
0V	5V	Off	On	Turning left
5V	0V	On	Off	Turning right
5V	5V	On	On	Straight

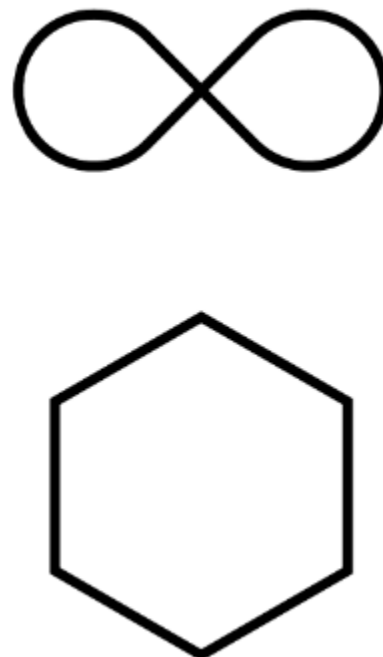
For testing the IR sensors combined with the Arduino, the code would print strings to the terminal depending on what the code was receiving from the sensors. Then changing the color of what the two IR sensors saw, seeing if the IR sensor's result LED outputs would lead to the correct terminal prints. If the Arduino saw the two IR sensors detect white, it would print "forward". If the Arduino saw the left IR sensor detect black and the right IR sensor detect white, it would print "Left turn". If the left IR sensor detected white and the right IR sensor detected black, the Arduino would print "Right turn". For the case where both IR sensors detected black, the Arduino would print "black". Table 4 shows the code Arduino agreeing with the IR sensors.

Table 4: IR sensors and Arduino testing results table

Left IR sensor	Left result LED	Right IR sensor	Right result LED	Terminal print
White	On	White	On	forward
Black	Off	White	On	Left turn
White	On	Black	Off	Right turn
Black	Off	Black	Off	black

The last part of the testing was the fully assembled line follower robot. The tests consisted of many small, custom paths made of black tape, and the two paths, the hexagon and the infinity symbol, in the IED workshop. The custom test paths data was collected on were a straight line, a curved turn, a 90° turn, a squiggle, the hexagon, and the infinity symbol.

Figure 12: Custom test paths(Left), IED workshop paths(Right)



When testing each path, Table 5 shows the line follower robot's consistency in completing a path by recording how many times the line follower robot would complete the path out of at least 3 attempts.

Table 5: Assembled line follower robot consistency testing table

	Straight line	Curved turn	90° turn	Squiggle	Hexagon	Infinity symbol
Attempts	3	3	3	3	4	6
Successes	3	3	3	3	4	0
Success rate	100%	100%	100%	100%	100%	0%

The assembled line follower robot testing showed great consistency for all the cases that didn't have any prolonged turns of turn radius less than 13cm, or intersections. The infinity symbol had both and would prevent the car from completing the track once. Occasionally, the line follower robot would successfully pass the infinity path's intersection or its tight turn, but it was very inconsistent. Unfortunately, the completion time was not recorded for these tests because the priority when designing the line follower robot was accuracy instead of speed.

Conclusion

The line follower robot project successfully achieved its main objective of developing a cost-efficient car capable of following a black line on a white background. During the design process, the team evaluated different sensing and control options. For the final design, the group decided on the L298N motor driver and IR sensors due to their favorable balance of cost, ease of implementation, and performance. The Arduino microcontroller was used as the control unit, as it was readily available and easy to use, especially for educational purposes.

Initial testing revealed issues with the positioning of the IR sensors, leading to inaccuracies in detecting the black line. The problem was resolved by adjusting the sensor placement to ensure direct contact with the ground, significantly improving tracking performance. The project demonstrated that careful concept generation and iterative design adjustments were crucial for success.

This work emphasizes the importance of balancing cost, reliability, and simplicity in engineering design, especially in projects intended for educational use. The final product is an accessible and adaptable product that may provide students with practical, hands-on experience in electronics and embedded systems.

Lessons Learned

Working on this project as part of a team provided valuable insights into various non-technical aspects that are critical for successful teamwork. One of the most essential lessons learned was the significance of effective communication among team members. Establishing a group chat ensured that everyone understood their deadlines and what needed to be done. Another important lesson learned was to start earlier, procrastination and laziness lead to a low start to the testing and improvement stage, and starting early could have drastically improved the competition scores and everything wouldn't have felt rushed. Overall, this project underscored the importance of strong communication which is critical to engineering success.

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Appendix - Software / Technology Used

Collaboration Among Team Members

- Discord
- Text Messages
- Webex Space

Design

- iPad Drawings
- Microsoft Word: For drafting and finalizing the project memo.
- Google Docs: Collaborative editing and drafting of documentation

- Excel: Used for organizing the concept selection matrix and analyzing design criteria

Programming

- Arduino IDE
- VScode

Testing/Simulation/Emulation

- Arduino IDE